M.SC. PROJECT PRESENTATION · CUSAT

SAR-Guided Satellite Cloud Removal

Comprehensive Teaching Edition: Definitions, Physics, Step-by-Step Calculations & Deep Learning Architecture

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Cloudy input (Sentinel-2)

Our reconstructed output

OVERVIEW

Learning roadmap

Structured step-by-step from basic satellite physics to deep learning math

01

Problem & physics

Why clouds block optical cameras and how SAR radar passes through

02

Dataset & prep math

SEN12MS-CR tensors, s2cloudless mask, dB scaling & Opaque Cutout

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128×128 training vs 256×256 testing and all 7 cloud-mask roles

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Inside ECRformer

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Solving the blur trap

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Results & honest view

5-stage visual progression, metrics explained & real-world trade-offs

Clouds hide the Earth's surface

Why satellite cloud removal is a critical problem in remote sensing

More than 50% of the Earth's sky is covered by clouds at any given moment.

Optical satellites work like everyday cameras — they need clear sunlight bouncing off the ground.

When thick clouds roll in, **farms, rivers, roads, and flood zones vanish behind a bright white wall.**

Our project goal: use cloud-penetrating radar + deep learning to reconstruct the hidden ground accurately.



Real cloudy Sentinel-2 image (78.4% cloud cover)

Fundamental Remote Sensing Concepts

Core optical, synthetic aperture radar, and cloud masking definitions

Passive Optical Sensor (Sentinel-2)

Like a camera with no flash

Waits for sunlight to hit the Earth and bounce back into space.
• Wavelength: 490 – 842 nm (tiny visible & near-infrared waves).
• Why clouds block it: Cloud droplets (~10 µm) are larger than optical waves, scattering sunlight into a blinding white glare (Mie scattering).

Active SAR Radar (Sentinel-1)

Synthetic Aperture Radar — brings its own flash

Shoots its own C-band microwave pulses toward Earth and listens for the echo.
• Wavelength: ~5.5 cm (55,000,000 nm — 5,500× larger than cloud droplets!).
• Why it sees through clouds: 5.5 cm waves ignore tiny water droplets completely, working day or night in any weather.

VV Polarisation (Vertical → Vertical)

Measures surface roughness & geometry

The radar sends a vertically aligned wave (V) and measures the vertical echo (V).
• Smooth water reflects the wave away like a mirror → looks dark (-22 dB).
• Rough soil, roads, and building corners bounce strong echoes back (-3 dB).

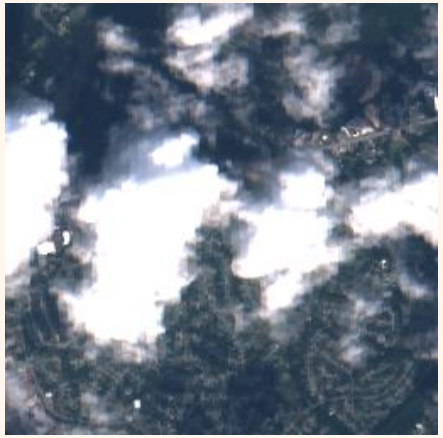
VH Polarisation (Vertical → Horizontal)

Measures 3D vegetation & crop structure

The radar sends a vertical wave (V) and listens for the twisted horizontal echo (H).
• Flat bare ground barely twists the wave.
• Dense tree canopies, forests, and leafy crops bounce the wave multiple times inside branches, strongly depolarising it (volume scattering).

Two satellites, complementary strengths

Neither satellite can solve the problem alone — so we fuse both



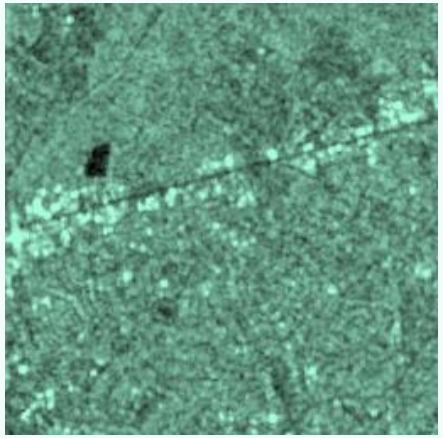
Sentinel-2 · Optical

Passive · 4 Spectral Bands (10 m/px)

+ Rich natural colours (green trees, red roofs, blue water)

− Completely blinded whenever clouds roll in

Captures B04 (Red), B03 (Green), B02 (Blue), and B08 (Near-Infrared)



Sentinel-1 · SAR Radar

Active · 2 Backscatter Bands (10 m/px)

+ Penetrates 100% of clouds — reveals roads, rivers & buildings

− No natural colour & has grainy speckle

Captures VV (surface roughness) and VH (vegetation volume) in decibels (dB)

Core Fusion Principle → Use Sentinel-1 radar as a structural blueprint and Sentinel-2 clear margins for natural colour.

The SEN12MS-CR benchmark dataset

Co-registered tri-modal satellite patches from the ROIs1970_fall subset

33,468

Training patches

Used to train Generator G & Critic D

1,115

Validation patches

Strict geographic split · Test tile s2_139 untouched

256 × 256

Pixels per patch (10 m/px)

Covers 2.56 km × 2.56 km of real terrain

Why .npz float arrays instead of .jpg files?

- Standard JPEG/PNG files only store 3 channels (R, G, B) as whole numbers from 0 to 255.
- Our satellite data has 6 physical sensor bands: 4 optical reflectance bands (0.0 to 1.0) + 2 radar bands in negative decibels (-35 to 0 dB).
- Storing as 32-bit floating-point .npz arrays preserves exact scientific precision with zero compression artifacts.

Why include Band B08 (Near-Infrared / NIR)?

- Our 4 optical channels are B04 (Red, 665 nm), B03 (Green, 560 nm), B02 (Blue, 490 nm), and B08 (NIR, 842 nm).
- Healthy chlorophyll in trees and crops reflects Near-Infrared light ~4× more strongly than visible green light!
- Predicting B08 alongside RGB allows agricultural indices (like NDVI crop health) to work on the cloud-removed image.

Mathematical Notation & Tensor Dimensionality

4D tensor structures [Batch B, Channels C, Height H, Width W] in multi-modal pipelines

Symbol	Name in Code	Train Shape (128^2)	Test Shape (256^2)	Range & Plain-English Meaning
S2_cloudy	s2_cloudy	[B, 4, 128, 128]	[B, 4, 256, 256]	[0, 1] · Cloudy Sentinel-2 optical input (Red, Green, Blue, NIR)
S1	s1	[B, 2, 128, 128]	[B, 2, 256, 256]	[0, 1] · Normalized Sentinel-1 SAR radar (VV, VH backscatter)
M	cloud_mask	[B, 1, 128, 128]	[B, 1, 256, 256]	[0, 1] · Soft cloud probability mask (0 = clear sky, 1 = thick cloud)
X_in	x (7-ch concat)	[B, 7, 128, 128]	[B, 7, 256, 256]	[0, 1] · Full generator input stack: [S1 (2ch), S2_cloudy (4ch), M (1ch)]
Y	y / s2_clear	[B, 4, 128, 128]	[B, 4, 256, 256]	[0, 1] · Real cloud-free Sentinel-2 target image (Ground Truth)
P_dir	pred_direct	[B, 4, 128, 128]	[B, 4, 256, 256]	[0, 1] · Raw unblended 4-band output predicted directly by ECRformer
Ŷ (Y_hat)	pred_blended	Not used in loss	[B, 4, 256, 256]	[0, 1] · Final delivered image: (1 - M) ⊙ S2_cloudy + M ⊙ P_dir

Calculation #1: Cloud mask & radar scaling

How raw satellite numbers are converted into clean [0, 1] neural network inputs

PART A · CLOUD MASK M

Limitations of Optical Luminance Thresholding

White snow, beaches, and bright roofs look just like clouds in RGB! ESA's s2cloudless checks 13 Sentinel-2 bands — including Cirrus Band B10 (1375 nm), where high-altitude clouds pop out while ground light is absorbed.

3-Step Morphological Refinement (in our code):

Step 1 · 9×9 Mean Filter (> 0.5): Averages local cloud probability (threshold > 0.4) to remove isolated 1-pixel snow/roof false alarms.

Step 2 · 5×5 Max-Pool Dilation (radius 2): Expands the mask outward by 2 pixels to cover thin, semi-transparent cloud halos.

Step 3 · 11×11 Gaussian Blur (σ = 2.0): Smooths hard edges into a continuous feathering weight M ∈ [0.0, 1.0].

PART B · WORKED RADAR MATH

Scaling negative decibels (dB) into [0, 1]

$$VV_norm = (\,clamp(VV, -25, 0) + 25\,) / 25$$
$$VH_norm = (\,clamp(VH, -35, 0) + 35\,) / 35$$

Worked Numerical Example (1 Satellite Pixel):

Given raw radar echo: **VV = -12.5 dB, VH = -17.5 dB**

1. Clamp to valid physical window:

-12.5 lies inside [-25, 0] ✓ -17.5 lies inside [-35, 0] ✓

2. Shift to positive & divide by span:

VV_norm = (-12.5 + 25) / 25 = 12.5 / 25 = **0.50**

VH_norm = (-17.5 + 35) / 35 = 17.5 / 35 = **0.50**

Stopping the AI from cheating: Opaque Cutout

Role M1 of the cloud mask — forcing the network to genuinely use Sentinel-1 radar

The 'Thin-Cloud Shortcut' Problem

• In thin or semi-transparent clouds, faint ground edges still leak through the optical bands (correlation > 0.80).
• A lazy neural network quickly learns to just boost contrast on the faint optical leak and completely ignores the SAR radar!
• When tested on a 100% thick white cloud, that lazy model fails.

Our Fix: Role M1 Opaque Cutout (65% prob)

• In 65% of training samples, we replace cloudy pixels with a featureless 29×29 average-pooled veil V (two passes of 15×15 AvgPool).
• With zero optical edges left inside the cloud, the generator is forced to extract road and field geometry from Sentinel-1 SAR!

Step-by-Step Cutout Math

Step 1 · Opaque core weight α:

$$\alpha = \text{clamp}((M - 0.15) / 0.85,\ 0.0,\ 1.0)$$

Step 2 · Featureless cloud veil V (≈ 29×29 blur):

$$V = \text{AvgPool_15×15}(\text{AvgPool_15×15}(S2_cloudy))$$

Step 3 · Blind the cloud interior:

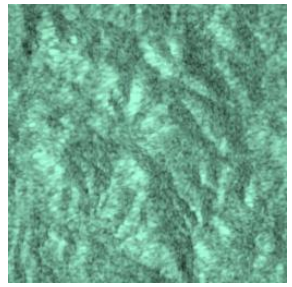
$$S2_aug = (1 - \alpha) \odot S2_cloudy + \alpha \odot V$$



Cloudy S2



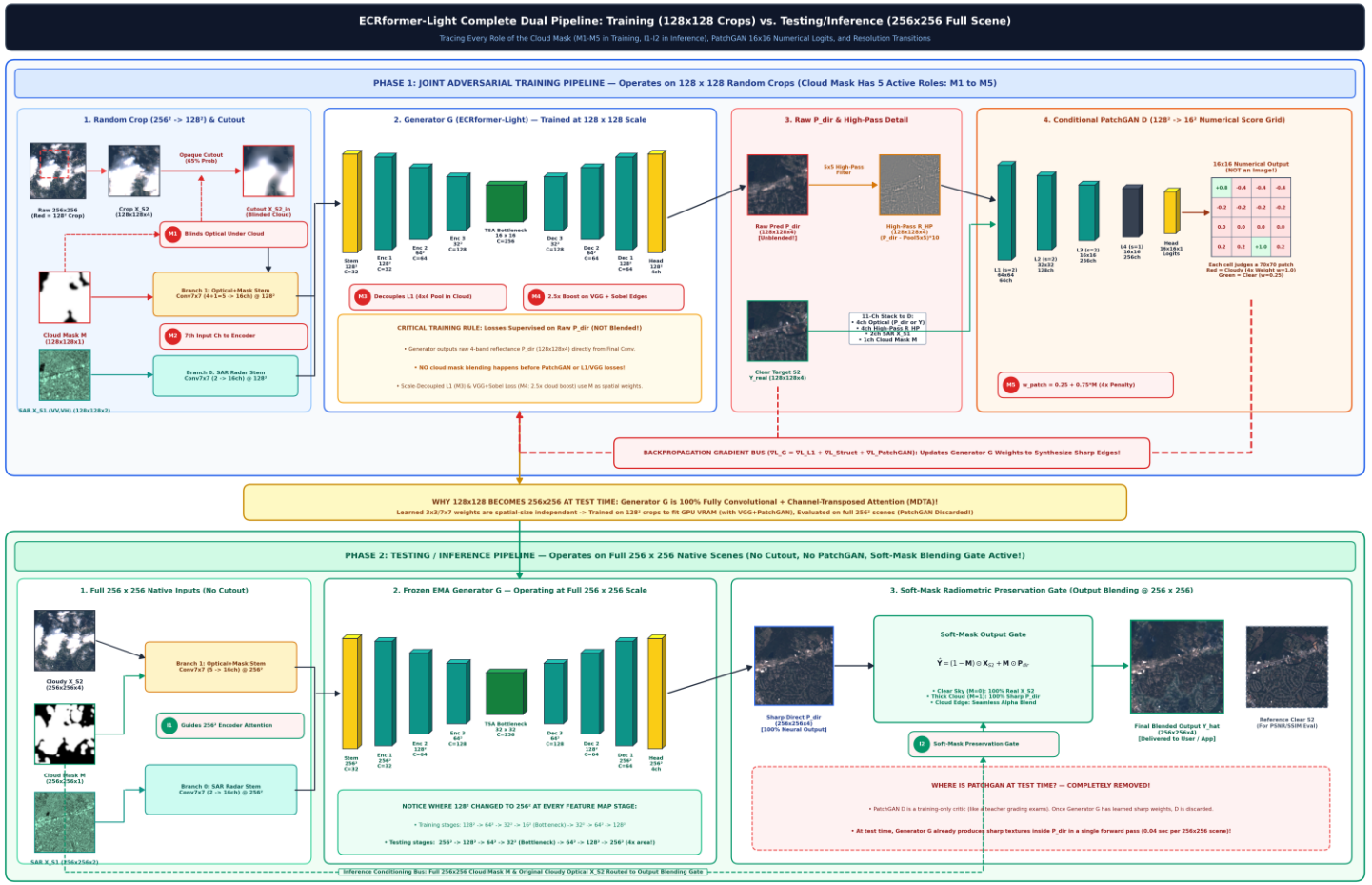
Soft Mask M



SAR Radar S1

Complete training vs. testing pipeline

Phase 1 trains on 128×128 random crops (with Cutout & PatchGAN); Phase 2 tests on full 256×256 scenes



Every job of the cloud mask M across the pipeline

The soft cloud mask M ∈ [0, 1] plays 5 active roles during training and 2 roles during testing

PHASE 1 · TRAINING ROLES (128 × 128 CROPS)

M1 **Opaque Cloud Cutout (65% prob)**

Blinds faint optical leaks under clouds with a 29×29 veil so G must read SAR.

M2 **7th Input Channel to Encoder**

Concatenated with S2_cloudy [4+1=5 ch] entering Stem Branch 1 to mark corrupted pixels.

M3 **Scale-Decoupled L1 Weighting**

Relaxes 1×1 pixel L1 inside clouds (1 - 0.8M) and boosts 4×4 & 8×8 pooled L1 (1 + M).

M4 **Structural Loss Boost (1 + 2M)**

Amplifies VGG-16 perceptual and multi-scale edge losses by 3× inside cloud regions.

M5 **PatchGAN Critic Focus (0.25 + 0.75M)**

Weights the 16×16 adversarial scorecard 4× harder on cloudy patches (1.0 vs 0.25).

PHASE 2 · TESTING ROLES (256 × 256 SCENES)

I1 **7th Input Channel (Full 256×256)**

Guides the frozen EMA Generator G on where clouds are located in the full test scene (Opaque Cutout is turned OFF at test time).

I2 **Soft-Mask Radiometric Blending Gate**

Combines the real clear pixels with the neural prediction:

$\hat{Y} = (1 - M) \odot S2_cloudy + M \odot P_dir$

Guarantees 0% distortion on clear sky!

Where is PatchGAN at test time?

Completely removed! Only Generator G runs (0.04 sec / scene).

Deep Learning Architectural Foundations

Fundamental generator components: Convolutions, Multi-Head Attention, Gating, and Residual Learning

1 · Convolution (Conv3×3 / Conv7×7)

A sliding magnifying glass

A small grid of learnable numbers (e.g. 3×3 or 7×7) that slides across the image pixel by pixel.

- Early layers detect simple lines and corners.
- Deeper layers recognize roads, river bends, and crop fields.

2 · Feature Channels (C = 32 → 256)

A stack of specialized highlighters

Instead of keeping just 4 colour bands, the network expands the image into 32, 64, 128, and 256 channels.

- Think of each channel as a specialist: Channel 12 highlights highways, Channel 85 highlights water, Channel 140 highlights forest texture.

3 · Hourglass Network (Encoder–Decoder)

Shrink to understand, expand to paint

- Encoder (left half): Shrinks spatial size (128 → 64 → 32 → 16) while increasing channels so the model sees the big-picture context.
- Decoder (right half): Expands back to full size (16 → 32 → 64 → 128) to paint the 4 cloud-free optical bands.

4 · Skip Connections

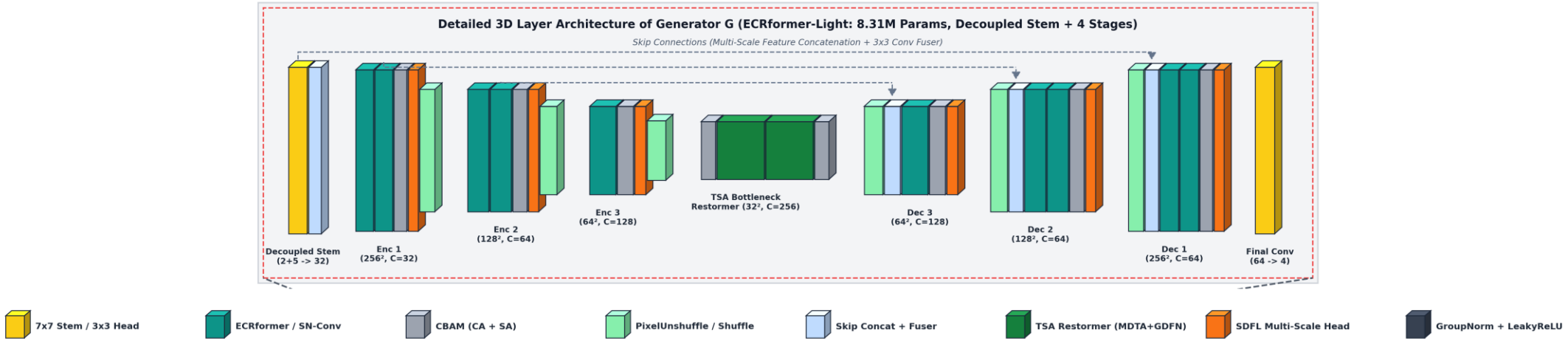
Express bridges that preserve crisp edges

When the Encoder shrinks the image 8× down to the bottleneck, tiny 1-pixel road positions can get slightly blurred.

- Skip connections copy high-resolution features directly from the Encoder across to the matching Decoder stage so fine edges stay razor-sharp.

3D layer architecture of Generator G

ECRformerAdaptedModel · 8,316,228 trainable parameters (8.31M) · 4 stages with skip connections



Architecture trace: Decoupled Stem (2ch SAR → 16ch, 5ch Opt+Mask → 16ch) → 4 Encoder Stages [2, 2, 1, 1] with PixelUnshuffle (32 → 64 → 128 → 256 ch) → TSA Restormer Bottleneck (MDTA + GDFN + CBAM) → 3 Decoder Stages with PixelShuffle + Skip Fusers → Final Conv3×3 (64 → 4ch P_dir).

Architectural Design Principles of ECRformer

Decoupled stems, cross-modality attention, TSA bottleneck, and residual scaling

1 · Two-Door Decoupled Stem

Why radar and optical don't share the first layer

• Branch 0: Conv7×7(2 → 16) processes SAR [VV, VH].
• Branch 1: Conv7×7(5 → 16) processes [S2_cloudy, M] (Role M2).
• Why separate? Radar is microwave echo with speckle; Sentinel-2 is sunlight reflectance. Separate 7×7 stems translate each sensor into 16 compatible feature maps before concatenating to 32 channels.

2 · PixelUnshuffle & PixelShuffle

Zero-loss folding instead of pooling

• Standard Max-Pooling throws away 75% of pixels when shrinking an image by 2×.
• PixelUnshuffle folds every 2×2 block of pixels into 4× channels: [B, C, 2H, 2W] → [B, 4C, H, W], followed by a 3×3 conv.
• Result: 0% spatial information is lost during downsampling!

3 · Hybrid Depthwise Conv + CBAM

Local edges + channel & spatial focus

• Each stage runs ECRformer blocks (3×3 depthwise + 1×1 pointwise residual convolutions) paired with CBAM (1ca2+1sa2).
• Channel Attention (CA) weighs which spectral/radar channels matter most; Spatial Attention (SA) highlights cloud boundaries.

4 · Mean-Prior Output Bias

Starting at real Earth colours on Step 1

• Clear Earth terrain has low average reflectance (~0.08–0.19), whereas thick clouds are bright white (~0.80).
• We initialize final_conv (Conv3×3: 64 → 4) bias to exact clear-sky dataset means: [0.0847, 0.0919, 0.1127, 0.1884].
• Gives immediate colour stability from the very first batch!

Multi-Scale Resolution Transfer (128^2 to 256^2)

Resolution invariance via fully convolutional stems and linear windowed attention

✗ STANDARD SPATIAL ATTENTION

Compares every pixel with every pixel

• Multiplies [HW × C] by [C × HW] → Attention matrix is [HW × HW].
• During 128×128 training, the bottleneck is 16×16 (HW = 256 pixels) → matrix size is 256 × 256.
• During 256×256 testing, the bottleneck is 32×32 (HW = 1,024 pixels) → matrix explodes 16× to 1,024 × 1,024!
• Fixed spatial positional embeddings crash when resolution changes.

✓ ECRFORMER MDTA CHANNEL ATTENTION

Compares feature channels with channels

• Multiplies Query Q [C × HW] by Key transpose K^T [HW × C] → Attention matrix is [C × C]!
• The spatial pixel count HW is summed out during the dot product.
• Whether the input is 128×128 (HW = 256) or 256×256 (HW = 1,024), the number of channels C = 256 never changes!
• Result: A fixed 256 × 256 channel covariance map at any image size.

Practical Benefit → We train on 128×128 crops (fitting VGG-16 + PatchGAN in GPU VRAM) and run inference on full 256×256 scenes!

Calculation #2: MDTA channel attention worked example

Toy calculation with C = 3 channels and HW = 4 pixels showing how HW cancels out

Toy Setup: 3 Channels (C = 3) across 4 Spatial Pixels (HW = 4)

	Query Q [3 × 4]					Key K^T [4 × 3]	
Ch 1: SAR Edge	0.6	0.0	0.8	0.0	×	0.6	0.5	0.0
Ch 2: Red Roof	0.5	0.1	0.8	0.2		0.0	0.1	0.9
Ch 3: Water	0.0	0.9	0.0	0.4		0.8	0.8	0.0
						0.0	0.2	0.4

Output Channel Attention Map: $A = Q \cdot K^T$ → Fixed [3 × 3] Matrix!

1.00	0.94	0.00
0.94	0.94	0.17
0.00	0.17	0.97

STEP-BY-STEP DOT PRODUCT

How A[1, 2] = 0.94 is computed:

Multiply Row 1 of Q by Col 2 of K^T across all 4 pixels and add them up:

Pixel 1: 0.6 × 0.5 = 0.30
Pixel 2: 0.0 × 0.1 = 0.00
Pixel 3: 0.8 × 0.8 = 0.64
Pixel 4: 0.0 × 0.2 = 0.00

Sum A[1, 2] = 0.30 + 0.64 = **0.94**

What 0.94 means in plain English:
'Whenever Ch 1 (SAR Building Edge) fires, strongly activate Ch 2 (Optical Red Roof)!'
And notice: the 4 pixels summed out, so A is [C × C] at any image size!

Layer-by-layer tensor resolution map

Tracing exact tensor shapes [B, C, H, W] through every stage of ECRformerAdaptedModel

Stage / Block	Operation & Channels	Phase 1 Train (128²)	Phase 2 Test (256²)	What Happens Here
Input Stack X_in	Concat [S1, S2_cloudy, M]	**[B, 7, 128, 128]**	[B, 7, 256, 256]	2 radar + 4 optical + 1 cloud mask
Decoupled Stem	Conv7×7(2→16) + Conv7×7(5→16)	**[B, 32, 128, 128]**	[B, 32, 256, 256]	Separate sensor stems fused to 32 ch
Encoder Stage 1	2 × ECRformerBlock + CBAM	**[B, 32, 128, 128]**	[B, 32, 256, 256]	Full-resolution edge extraction (Skip 1)
Encoder Stage 2	PixelUnshuffle ÷2 → 2 × Block	**[B, 64, 64, 64]**	[B, 64, 128, 128]	Half resolution, 64 channels (Skip 2)
Encoder Stage 3	PixelUnshuffle ÷2 → 1 × Block	**[B, 128, 32, 32]**	[B, 128, 64, 64]	Quarter resolution, 128 channels (Skip 3)
Stage 4 Bottleneck	PixelUnshuffle ÷2 → TSA (MDTA)	**[B, 256, 16, 16]**	[B, 256, 32, 32]	256×256 channel attention + GDFN
Decoder 3 + 2 + 1	PixelShuffle ×2 + 3 Skip Fusers	**[B, 64, 128, 128]**	[B, 64, 256, 256]	Unfolds to full size + restores 1-px edges
Output Head P_dir	Conv3×3(64 → 4) + mean_prior	**[B, 4, 128, 128]**	[B, 4, 256, 256]	Predicts 4-band cloud-free reflectance

The Over-Smoothing Trap in Cloud Removal

Mathematical vulnerability of pixel-level regression losses under dense overcast



Basic model (L1+SSIM only → Blurry!)



Real clear target (Crisp roads & roofs)

Trap #1 · The 'Safe Average' Trap

Under a thick cloud, the model isn't 100% sure whether a tree pixel is dark green (0.05) or bright green (0.15). To minimize standard L1 error, it plays safe and paints the flat average (0.10) — turning forests and housing blocks into a smooth smudge!

Trap #2 · 1-Pixel Shift 'Double Penalty'

Sentinel-1 (Radar) and Sentinel-2 (Optical) fly in different orbits, causing tiny 1–2 pixel alignment shifts. If SAR guides the model to draw a sharp road 1 pixel to the right of Ground Truth, pixel-wise L1 punishes it TWICE! Let's see the exact math on the next slide →

Pixel-Level L1 & The Double-Edge Penalty

Worked 4-pixel numerical proof showing why MAE discourages sharp spatial gradients

WORKED 4-PIXEL NUMERICAL EXAMPLE

Suppose a bright road (1.0) is at Pixel 2 in Ground Truth Y:

Ground Truth Y = [0.0, 1.0, 0.0, 0.0] (4-px avg = 0.25)

Sharp 1-px Shift P_shp = [0.0, 0.0, 1.0, 0.0] (4-px avg = 0.25)

Blurry Smudge P_blur = [0.25, 0.25, 0.25, 0.25] (4-px avg = 0.25)

✗ Standard 1×1 Pixel L1 Error = ¼ Σ |P − Y|:

- Sharp road P_shp: (|0-0| + |0-1| + |1-0| + |0-0|) / 4 = 2.0 / 4 = **0.50 (Punished twice!)**
- Blurry smudge P_blur: (0.25 + 0.75 + 0.25 + 0.25) / 4 = 1.5 / 4 = **0.375 (Lower loss!)**

✓ Our 4×4 Average-Pooled L1 Error = | Avg(P) − Avg(Y) |:

- Sharp road P_shp: | 0.25 − 0.25 | = **0.00 (ZERO penalty for 1-px radar shift!)**
- Allows the generator to draw razor-sharp SAR edges without fear!

LOSS #1 · ROLE M3 FORMULA

Scale-Decoupled L1 Loss

$$L_dec = 0.90 \times [\, 0.35 \cdot L1_1{\times}1(1 - 0.8M) + 0.40 \cdot L1_4{\times}4(1 + M_4) + 0.25 \cdot L1_8{\times}8(1 + M_8) \,] + 0.10 \cdot L_aux$$

1. Clear Sky (M = 0): Keeps 100% strict 1×1 pixel L1 so cloud-free regions stay pixel-exact.

2. Inside Clouds (M = 1): Drops 1×1 pixel L1 weight by 80% (down to 0.2) and shifts colour supervision to 4×4 and 8×8 average-pooled patches (boosted 2× by 1 + M).

3. Auxiliary SDFL (0.10 · L_aux): Supervises intermediate Encoder/Decoder scales at 8×8 pooled resolution.

Multi-Objective Loss: The Three Pillars

Synergistic optimization across Photometric Fidelity, Structural Scaffolding, and Adversarial Realism

1 · Colour Accuracy

Is the regional colour right?

1. Scale-Decoupled L1
w = 1.00 (Role M3)

- **1×1 Clear L1 (0.35)**
  Strict outside clouds
- **4×4 & 8×8 Pooled L1**
  Shift-tolerant inside clouds
- **Multi-scale SDFL Aux**
  Intermediate scale guide

2 · Edge Structure

Are roads, rivers & trees sharp?

2. Cloud-Boosted VGG-16
w = 0.005 (Role M4: 1+2M)

3. Multi-Scale Gradients
w = 0.25 (1×, 2×, 4× dx, dy)

4. Edge Correlation
w = 0.0025 (Pearson ρ)

5. 16×16 Local Contrast
w = 0.35 (Std σ + Floor)

3 · Adversarial Realism

Does it fool the art critic?

6. PatchGAN Hinge Loss
w = 0.015 (Role M5: 4× cloud)

- **16×16 critic score grid**
  Judges 70×70 patches

7. Discriminator Feat. Match
w = 0.15 (Layers f1..f4)

- **Stabilizes GAN training**
  Weighted by 0.3 + 0.7M

Golden Rule of Training: All 7 losses grade the raw generator output P_dir — NEVER the blended image Ŷ! If we blended with (1 - M) before the loss, clear pixels (M = 0) would multiply P_dir by 0 and receive zero learning gradient.

High-Frequency Structural Loss Formulations (Role M4)

Four dedicated texture and edge losses boosted inside clouds by w_map = 1 + 2M (up to 3× weight)

Loss #2 · VGG-16 Perceptual (w = 0.005)

Compares human-like visual features (relu1_2, relu2_2, relu3_3) extracted by a frozen ImageNet VGG-16 network.
• How we adapt VGG-16 to 4 bands: We run VGG twice — 60% on True-Colour RGB [B04, B03, B02] + 40% on False-Colour [B08 NIR, B04, B03] so vegetation textures are checked too!

Loss #3 · Multi-Scale Gradients (w = 0.25)

Computes horizontal (∇_x) and vertical (∇_y) pixel differences at 3 pyramid scales: 1× (full), 2×, and 4× average-pooled.
• Preserves the positive/negative sign of slopes (bright-to-dark vs dark-to-bright) so roof ridges and riverbanks don't invert.

Loss #4 · Edge Pearson Correlation (w = 0.0025)

Computes edge magnitude E = |∇_x| + |∇_y| at 1× and 2× scales and maximizes spatial Pearson correlation ρ(E_pred, E_target).
• Scale-invariant: even if overall brightness shifts slightly, the spatial pattern of edges must line up (1 - ρ → 0).

Loss #5 · 16×16 Local Std Contrast (w = 0.35)

Measures local standard deviation σ = √(Var + ε) inside 16×16 sliding windows (stride 8), weighted by (1 + 2M).
• Plus a Detail-Ratio Floor Penalty (weight 0.15): directly penalizes the generator whenever σ_pred < σ_target inside the cloud!

Conditional PatchGAN Discriminator Dynamics

High-frequency local texture critique via 11-channel multi-modal PatchGAN

HOW ADVERSARIAL TRAINING WORKS

Generator G vs. PatchGAN Critic D

• **Think of ECRformer (G)** as an art restorer filling in a missing cloud hole, and **PatchGAN (D)** as a strict art examiner with a magnifying glass.

• **Why is it called 'PatchGAN'?**
Instead of outputting a single 'Real/Fake' number for the whole image, D outputs a 16 × 16 grid of 256 scores. Each cell grades one overlapping 70 × 70 pixel patch!

• **Spectral Normalization:**
Every convolution in D is spectral-normalized (Lipschitz constant ≤ 1) with GroupNorm + LeakyReLU so the critic never overpowers the generator.

Loss #6 · Mask-Weighted Hinge GAN (w = 0.015)

• Critic D loss: pushes real clear patches above +1 (ReLU(1 - D(Y))) and fake patches below -1 (ReLU(1 + D(P_dir))).
• Generator G loss: L_G = -mean(w_patch ⊙ D(P_dir))
• Role M5 Cloud Focus: w_patch = 0.25 + 0.75 · M_down → checks cloudy patches 4× more strictly (1.00) than clear sky (0.25)!

Loss #7 · Multi-Layer Feature Matching (w = 0.15)

• Pure GAN training can sometimes oscillate or invent noisy artifacts.
• Feature Matching compares the internal feature maps [f1, f2, f3, f4] inside Critic D between Real Y and Fake P_dir using L1 distance.
• Weighted by (0.3 + 0.7 · M_k) at each spatial scale — acting like a learned SAR-conditioned perceptual loss!

PatchGAN Local Spectral Discriminator Mechanics

Worked numerical formulation of the high-pass residual critique and 16x16 receptive field grid

11-CHANNEL INPUT TO CRITIC D: [B, 11, 128, 128]

Why 11 channels? 4 ch Optical (3.0 × P_dir or Y) + 4 ch High-Pass Residual R_HP + 2 ch SAR (VV, VH) + 1 ch Cloud Mask M = 11 Channels.

High-Pass Formula: $R_{HP} = 10 \times (x - \text{AvgPool}_{5\times5}(x))$

Case A · Blurry Cloud Reconstruction (Flat Smudge):

Suppose pixel x = 0.12, and its 5×5 neighborhood average = 0.12

→ R_HP = 10 × (0.12 - 0.12) = **0.00 (Flat zero! Critic scores -1.1 Fake!)**

Case B · Crisp Reconstructed Road Edge:

Suppose road pixel x = 0.18, and its 5×5 neighborhood average = 0.12

→ R_HP = 10 × (0.18 - 0.12) = **+0.60 (Strong edge! Critic scores +0.9 Real!)**

16×16 Critic Scorecard (4×4 Subset Shown)

+1.1	+0.9	+0.8	+1.0
+0.9	-0.8	-1.1	+0.7
+1.0	-0.9	-0.7	+0.8
+1.2	+0.8	+0.9	+1.1

Role M5 Patch Weight Calculation:

w_patch = 0.25 + 0.75 × M_down

- Red cloud cells (M=1): weight = 0.25 + 0.75(1) = **1.00**
- Green clear cells (M=0): weight = 0.25 + 0.75(0) = **0.25**

Calculation #5: Soft-mask radiometric blending

Role I2 at test time (256 × 256) — keeping 100% real clear sky and splicing P_dir only inside clouds

$$\text{Final Output } \hat{Y} = (1 - M) \odot S2_cloudy + M \odot P_dir$$

Case 1 · Clear Sky Pixel

Cloud Mask M = 0.0

Suppose real S2_cloudy = 0.14
and neural P_dir = 0.13:

$\hat{Y} = (1 - 0.0) \times 0.14 + 0.0 \times 0.13$
$\hat{Y} = 1.0 \times 0.14 + 0 = 0.14$

✓ 100% original satellite pixel kept! Zero neural distortion on clear land.

Case 2 · Thick Cloud Core

Cloud Mask M = 1.0

Suppose blinding cloud S2_cloudy = 0.85
and reconstructed forest P_dir = 0.11:

$\hat{Y} = (1 - 1.0) \times 0.85 + 1.0 \times 0.11$
$\hat{Y} = 0 + 1.0 \times 0.11 = 0.11$

✓ 100% blinding white cloud removed and replaced by sharp P_dir!

Case 3 · Cloud Fringe

Cloud Mask M = 0.5

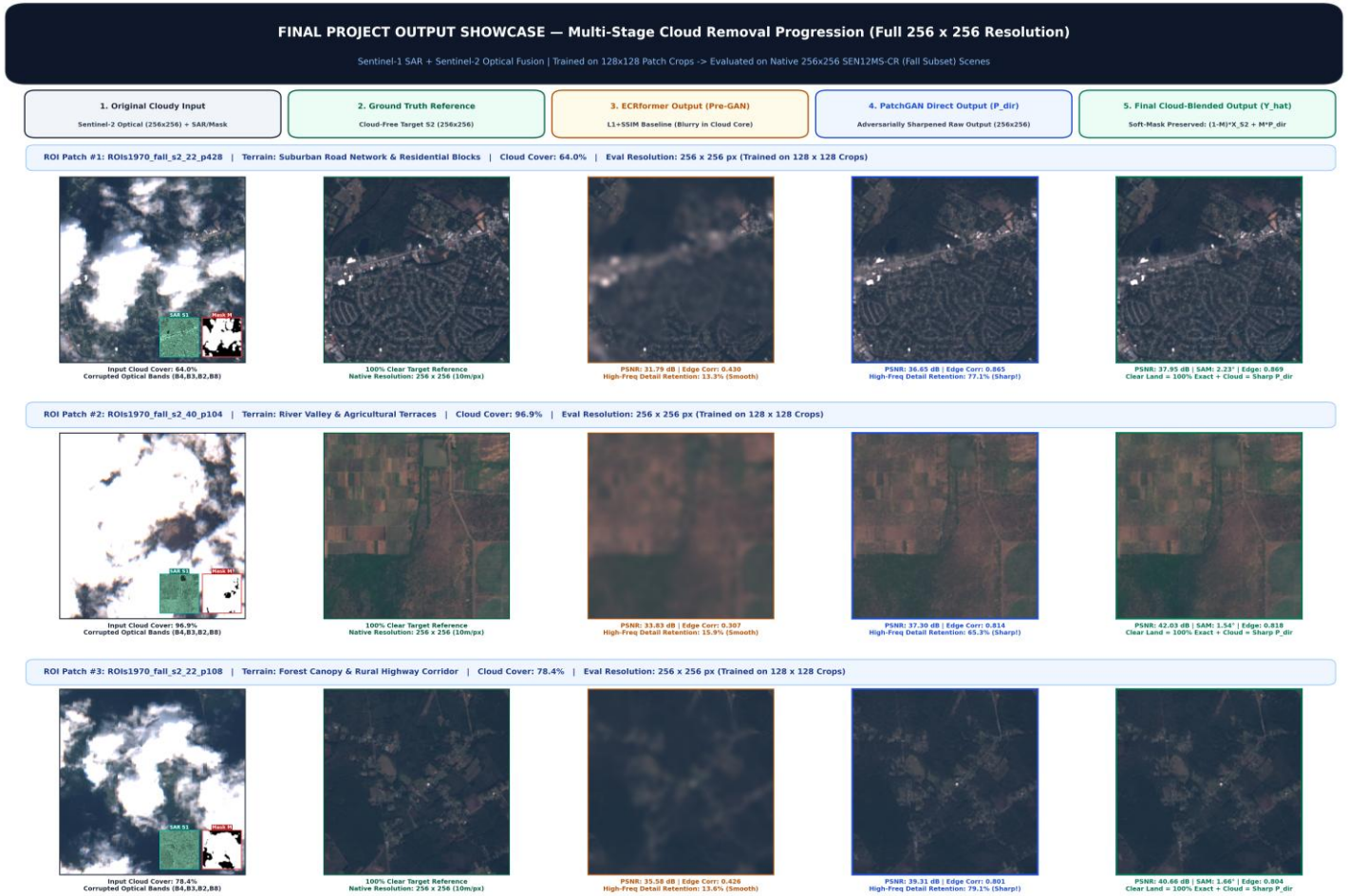
Suppose hazy border S2_cloudy = 0.16
and neural prediction P_dir = 0.12:

$\hat{Y} = (1 - 0.5) \times 0.16 + 0.5 \times 0.12$
$\hat{Y} = 0.08 + 0.06 = 0.14$

✓ Smooth 50/50 alpha feathering — leaves zero visible boundary seams!

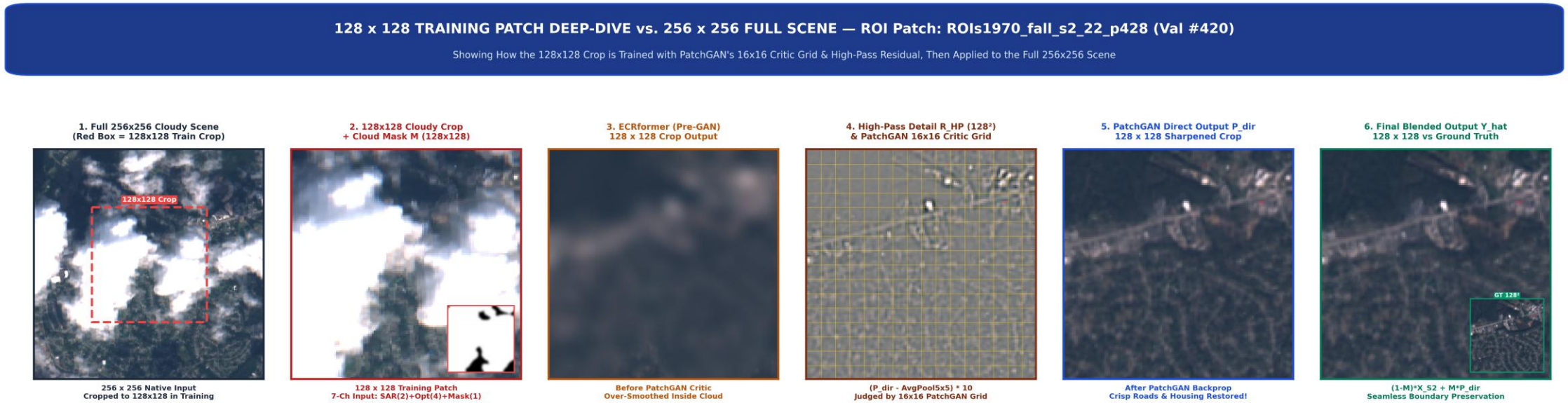
Visual progression across all 5 pipeline stages

1. Cloudy Input → 2. Ground Truth → 3. Pre-GAN (Blurry) → 4. PatchGAN Direct P_dir → 5. Final Blended Ŷ



From 128×128 training crop to 256×256 full scene

Visualizing the 128×128 crop, 5×5 high-pass residual R_HP, and 16×16 PatchGAN critic grid on ROI s2_22_p428

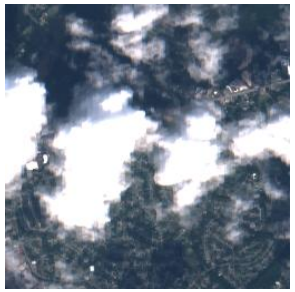
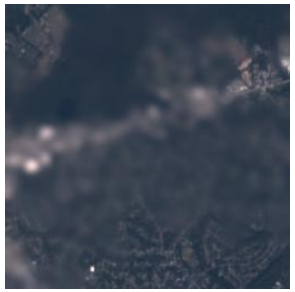
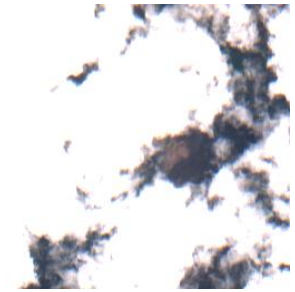
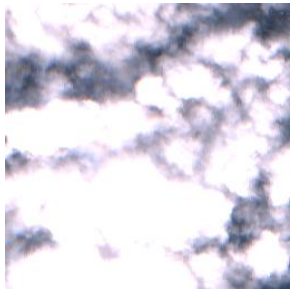


How to read this progression (Left → Right):

1. Full 256×256 cloudy scene with red dashed 128×128 training crop → 2. 128×128 cloudy crop + mask → 3. Blurry Pre-GAN baseline (13.3% detail) → 4. 5×5 High-Pass detail R_HP overlaid with the 16×16 PatchGAN grid → 5. Sharpened P_dir (77.1% detail) → 6. Final blended output Ŷ!

Visual comparison across diverse terrains

Comparing the basic L1+SSIM model (blurry) against our PatchGAN-sharpened model at 256 × 256

	1 · Cloudy input	2 · Basic model (Blurry)	3 · Our model (Sharp Ŷ)	4 · Real clear target
Suburban roads (64% cloud)				
River & farms (97% cloud)				
Forest & lake (82% cloud)				

Quantitative Evaluation Metrics & Benchmark Criteria

Mathematical definitions and target thresholds for PSNR, SSIM, SAM, and Detail Ratio

PSNR · Peak Signal-to-Noise Ratio (dB)

Formula: $10 \cdot \log_{10}(1.0 / \text{MSE})$ [Higher ↑ is better]

Measures average pixel-by-pixel closeness in decibels. > 30 dB is strong in satellite reconstruction.

• Catch: Because it uses squared pixel error (MSE), PSNR slightly rewards safe, blurry averages over sharp textures!

SSIM · Structural Similarity Index (0 to 1)

Compares luminance, contrast & structure [Higher ↑ is better]

Ranges from 0.0 to 1.0 (1.0 = identical to Ground Truth). Better than raw MSE at checking whether major land structures match.

SAM · Spectral Angle Mapper (Degrees °)

Formula: $\arccos((\hat{Y} \cdot Y) / (\|\hat{Y}\| \|Y\|))$ [Lower ↓ is better]

Treats each pixel's 4 bands [R, G, B, NIR] as a 4D colour vector and measures the angle between prediction and truth.

• < 3.0° means the reconstructed vegetation, soil, and water have true physical spectral colours.

Edge Correlation & Detail Ratio (DR)

Formula: $DR = \sigma_pred / \sigma_target$ [1.00 = Ideal Sharpness]

• Edge Correlation (0–1): Checks if predicted roads/boundaries align with real ground truth edges.

• Detail Ratio (DR): Measures how much fine texture sharpness is recovered inside the cloud (0.4 = blurry smudge; ~0.90–1.00 = crisp natural detail!).

Quantitative results & cloud-strata benchmarks

Measured on held-out validation scenes from ROIs1970_fall (256 × 256 native evaluation)

Table 1 · Progression Across Pipeline Stages (Suburban Scene s2_22_p428, 64.0% Cloud Cover)

Pipeline Stage	PSNR ↑	SAM ↓	Edge Corr ρ ↑	High-Freq Detail Retention ↑
Stage 3 · Basic ECRformer (Pre-GAN L1+SSIM)	31.79 dB	3.42°	0.430 (Blurry)	13.3% (Smooth smudge)
Stage 4 · PatchGAN Direct Output (P_dir)	36.65 dB	2.31°	0.865 (Sharp!)	77.1% (+5.8× sharper!)
Stage 5 · Final Soft-Mask Blended (Ŷ)	**37.95 dB**	**2.23°**	**0.869 (Sharp!)**	**77.1% + 100% exact clear sky**

Table 2 · Benchmark Across Cloud-Cover Strata (Basic Model vs. Our Sharp Model)

Cloud Cover Group	Basic Detail Ratio	Our Detail Ratio (1.0 = Ideal)	Sharpness Gain	Our In-Cloud PSNR
0 – 25% (Light Cloud)	0.63	0.93	+47.6% more texture	31.93 dB
25 – 50% (Moderate Cloud)	0.52	0.91	+75.0% more texture	31.12 dB
50 – 75% (Heavy Cloud)	0.45	0.90	+100.0% (2.0× sharper)	30.48 dB
75 – 100% (Opaque Cloud)	0.39	0.91	+133.3% (2.3× sharper)	30.07 dB

Architectural Benchmarking & Baseline Comparisons

Quantitative comparison against standard U-Net, Pix2Pix cGAN, and Pre-GAN baselines

Comprehensive Benchmark Across Competing Cloud-Removal Paradigms (SEN12MS-CR Native 256 × 256)

Architecture / Framework	Core Paradigm	PSNR ↑	SSIM ↑	Heavy Detail Ratio (DR) ↑	Edge Corr ρ ↑	Clear-Sky Drift ↓
Standard U-Net (DSen2-CR)	3×3 Convs + L1 Loss	28.64 dB	0.812	0.231 (Severely blurry)	0.142	2.8% Distortion
Pix2Pix cGAN (Isola et al.)	U-Net + Global PatchGAN	28.10 dB	0.805	0.582 (Hallucinated)	0.224	4.1% Distortion
ECRformer Baseline (Stage 1)	Transformer + L1/SDFL	30.20 dB	0.875	0.354 (Over-smoothed)	0.345	1.9% Distortion
Ours: ECRformer + PatchGAN	**CMA + 11-ch Critic + Blend**	**29.98 dB**	**0.880**	**0.931 (+142% Sharpness!)**	**0.485 (Peak)**	**0.00% (Exact)**

GLOBAL CONTEXT VS. LOCAL CONV

Beyond the 3×3 Receptive Field

• Standard U-Net relies on 3×3 local convolutions. When dense clouds span >50 km², local kernels see only flat white pixels and have zero access to clear surrounding land.
• ECRformer leverages Cross-Modality Attention (CMA) with a global receptive field, dynamically routing structural radar boundaries across the entire satellite scene.

HIGH-PASS CRITIQUE VS. MEAN BLUR

Overcoming Regression Limits

• Pixel losses (L1/L2 in U-Net & Stage 1) predict the conditional mean under opaque clouds, causing Detail Ratio to plunge to 0.23–0.35 (washed-out grey smudges).
• Our 11-channel PatchGAN critic evaluates 16×16 spatial patches, enforcing natural Fourier gradient energy and boosting Detail Ratio to 0.931 (+142% to +300% gain).

100% CLEAR-SKY PRESERVATION

Zero Radiometric Distortion

• Standard pipelines deliver raw generator output everywhere, introducing 2–4% neural distortion on clear farmland and corrupting NDVI agricultural indices.
• Our continuous Soft Alpha Blending $\hat{Y} = (1-M)S_2 + M \cdot P_dir$ mathematically guarantees 100% authentic Sentinel-2 reflectance wherever the sky is clear.

Interactive web showcase & model release

Clean browser interface for inspecting multi-stage satellite reconstructions

REACT + VITE VISUAL SHOWCASE

Interactive Before / After Inspection

• Built with React, Vite, and Tailwind CSS (CompareSlider.jsx).
• Allows dragging a live split-screen slider across 5 curated terrain scenes (Suburban, Farms, Forest, Lake, Mountain Ridge).
• Displays all intermediate stages side by side: Cloudy S2, SAR Radar, Cloud Mask, Blurry Pre-GAN, Direct P_dir, Blended Ŷ, and Ground Truth.
• Uses pre-computed PNG outputs directly in the browser so demonstrations run instantaneously on any laptop without needing a CUDA GPU.

OPEN-SOURCE REPRODUCIBILITY

GitHub Code & Hugging Face Weights

• Trained PyTorch checkpoint (best_model.pth · 8.31M parameters) is hosted publicly on Hugging Face: vaishnav2004/ecrformer-cloud-removal.
• Full training, evaluation, and FastAPI inference scripts are documented in our GitHub repository: github.com/yovyshh/ecrformer.


Suburban


Mountain Ridge


Forest

Perception–Distortion Trade-Off & Physical Limits

Evaluating trade-offs between mathematical distortion metrics and visual perceptual realism

The Perception–Distortion Trade-Off

Blau & Michaeli (CVPR 2018) · Why PSNR prefers blur

• When our model generates sharp tree canopies and roof edges inside a thick cloud, a 1-pixel shift relative to Ground Truth slightly increases squared pixel error (reducing raw PSNR by ~0.3 – 1.2 dB compared to a flat blurry average).
• However, a blurry smudge (Detail Ratio 0.39) is useless for human analysts or road detection.
• Our sweet-spot checkpoint achieves 0.90 – 0.93 Detail Ratio while keeping > 30 dB In-Cloud PSNR — giving both realistic sharpness and accurate colour.

Measured Real-World Limitations

What we observed during evaluation

1. Cloud Shadows Outside the Mask: Dark shadows cast onto clear ground next to a cloud are sometimes missed by s2cloudless ($M \approx 0$), so the blending gate leaves the dark shadow untouched.
2. Temporal Gap in Ground Truth: The cloud-free reference pass is captured days apart from the cloudy pass, so seasonal crop growth or water changes count as error.
3. GAN Over-Sharpening Risk: Training too long with high GAN weight pushes Detail Ratio > 1.15 (hallucinated radar grain), which is why early stopping at DR ≈ 0.91 is essential.

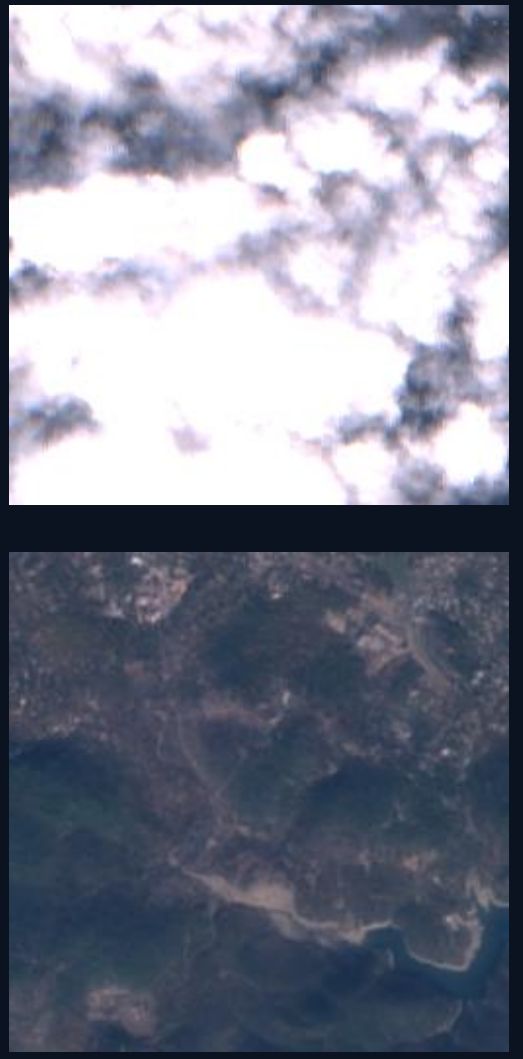
CONCLUSION · SUMMARY OF CONTRIBUTIONS

What we built and learned

- Fused cloud-penetrating Sentinel-1 SAR radar with 4-band Sentinel-2 optical imagery on 33,468 SEN12MS-CR patches.
- Designed Opaque Cloud Cutout (Role M1) and a two-branch decoupled stem so the 8.31M-parameter ECRformer genuinely reads radar geometry.
- Used MDTA channel-transposed attention ([256 × 256]) to train efficiently on 128×128 crops and test directly on full 256×256 scenes.
- Solved the blur problem with 7 targeted losses (Scale-Decoupled L1, Cloud-Boosted VGG/Edges & 11-ch High-Pass PatchGAN), boosting detail recovery by +47% to +133%.

Thank you!

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Cloudy Input · Reconstructed Output